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KINETICS OF BI-MOLECULAR REACTIONS

1- BIMOLECULAR COLLISION THEORY:

Bimolecular collision theory, for the reactions consisting of two reactant molecules only, was proposed by Max Trautz and William Lewis.

Postulates:

1- Reaction between two molecules takes place due to collision between them.

2- All collision between the molecules do not lead to a chemical reaction.

3- The collision which causes a chemical reaction to take place is termed as effective or successful collision, the other being ineffective.

4- For the collision to be effective the reactant molecules must have energies greater than the activation energy and proper orientation so as to cause head on collision.

5- The rate of reaction is directly proportional to the number of collisions per unit volume per unit time, the fraction of molecules with proper orientation and the fraction molecules having energies greater than the activation energy.

6- Variation in the rate of a reaction due to change in temp. or pressure is also due to fact that these factor changes the probability of the collision between the reactant molecules. Likewise a catalyst increases the rate of reaction just by lowering the activation energy. Hence, increasing the probability of effective collisions.

⇒ Mathematically:

In the light of postulates of collision theory, the rate of reaction is equal to the product of the number of collisions of the molecule A with the molecules B per unit volume per unit time, the fraction of the molecules having activation energy and the fraction of molecules with proper orientation.

$$\text{Rate} = (N_{AB})(e^{-E^*/RT})(P) \quad \text{——} (1)$$

In eq (1) "P" is called steric factor or probability factor. It is a dimensionless quantity. "N_{AB}" is called collision frequency which depend upon molar masses of reacting species (M_A and M_B), temp. of the medium (T), collision diameter (d), and molar concentration of reacting species (n_A and n_B).

$$N_{AB} = \left[8\pi RT\left(\frac{M_A + M_B}{M_A M_B}\right)\right]^{1/2} d^2 n_A n_B \text{ —} (2)$$

Putting the value of N_{AB} to eq (1) we get consider the reaction:

$$A + B \xrightarrow{k_2} P \quad \text{——} (3)$$

Rate of this reaction is given by

$$\text{Rate} = k_2 [A][B] \quad \text{——} (4)$$

$$\text{Rate} = k_2\, n_A\, n_B \quad \text{——} (5)$$

Where k_2 is the rate constant for the bimolecular reaction. n_A and n_B represent the number of molecules A and B respectively per unit volume.

By comparing eq ② and ⑤,

$$k_2 = Pd^2\left[\frac{8\pi RT(M_A+M_B)}{M_A M_B}\right]^{1/2} e^{-E^*/RT} \quad \text{——} ⑥$$

$$k_2 = DT^{1/2}\, e^{-E^*/RT} \quad \text{——} ⑦$$

where

$$D = Pd^2\left[\frac{8\pi R(M_A+M_B)}{M_A M_B}\right]^{1/2}$$

Equation ⑦ similar in form to the Arrhenius eq. shows that the rate constant for a bimolecular reaction increases exponentially with temp. but in the present case, the pre-exponential factor is a function of temperature.

A COMPARISON OF COLLISION THEORY WITH ARRHENIUS THEORY:

Taking ln on both sides of eq ⑦

$$\ln k_2 = \ln D + \frac{1}{2}\ln T - \frac{E^*}{RT}$$

$$\ln k_2 = \ln D - \frac{1}{2}\ln\left(\frac{1}{T}\right) - \frac{E^*}{R}\left(\frac{1}{T}\right)$$

Differentiating the above equation with respect to $1/T$, we get

$$\frac{d \ln k_2}{d(1/T)} = \frac{d}{d(1/T)}(\ln D) - \frac{1}{2}\frac{d}{d(1/T)}\ln\left(\frac{1}{T}\right) - \frac{E^*}{R}\frac{d}{d(1/T)}\left(\frac{1}{T}\right).$$

$$\frac{d \ln k_2}{d(1/T)} = 0 - \frac{1}{2}\left(\frac{1}{1/T}\right) - \frac{E^*}{R}(1)$$

$$\frac{d \ln k_2}{d(1/T)} = \frac{-1}{2}T - \frac{E^*}{R}$$

Multiplying the above eq by $-R$,

$$-R.\frac{d \ln k_2}{d(1/T)} = \frac{RT}{2} + E^*$$

or

$$\frac{RT}{2} + E^* = -R\frac{d \ln k_2}{d(1/T)} \quad \text{——— (8)}$$

Comparing eq (8) with Arrhenius eq,

$$E_a = -R\frac{d \ln k}{d(1/T)},$$ we get

$$E_a = E^* + \frac{RT}{2}$$

$$E^* = E_a - \frac{RT}{2}$$

Putting this value of E^* in eq (6), we get

$$k_2 = P\sigma^2\left\{\frac{8\pi RT(M_A+M_B)}{M_A M_B}\right\}^{1/2} e^{-[E_a - RT/2]/RT}.$$

$$k_2 = Pd^2\left[\frac{8\pi RT(M_A+M_B)}{M_A M_B}\right]^{1/2} e^{1/2} \cdot e^{Ea/RT} \quad \text{——— (9)}$$

Comparing eq (9) with Arrhenius eq ($K = Ae^{-Ea/RT}$) the expression for the pre-exponential factor can be obtained i.e.

$$A = Pd^2\left(\frac{8\pi RT(M_A+M_B)}{M_A M_B}\right)^{1/2} e^{1/2}$$

It is evident from the above eq. that the pre-exponential factor in collision theory, is a function of temperature.

⇒ Shortcomings of Collision Theory:

One of the weakness of Collision Theory is that it is applicable to simple gas phase reactions only and is not useful for many reactions in solution phase. This theory does not explain the cleavage and the formation of bond during a reaction. Collision theory does not explain surface catalyzed reactions. It is not applicable to complex chain reactions.

2- TRANSITION STATE THEORY:

This theory was formulated in 1935 by Henry Eyring.

Postulates :-

1- The reaction between two species say A and B leads to the formation of an activated complex.

2- Activated complex is highly unstable energetic specie that spontaneously decomposes to give the products.

3- The reaction between the reactants and the activated complex is reversible.

4- Activated complex is in dynamic equilibrium with reactants.

5- Formation of products from activated complex is an irreversible reaction.

6- Rate of formation of products depend upon the concentration of activation complex.

7- The concentration of activated complex is not a measurable quantity.

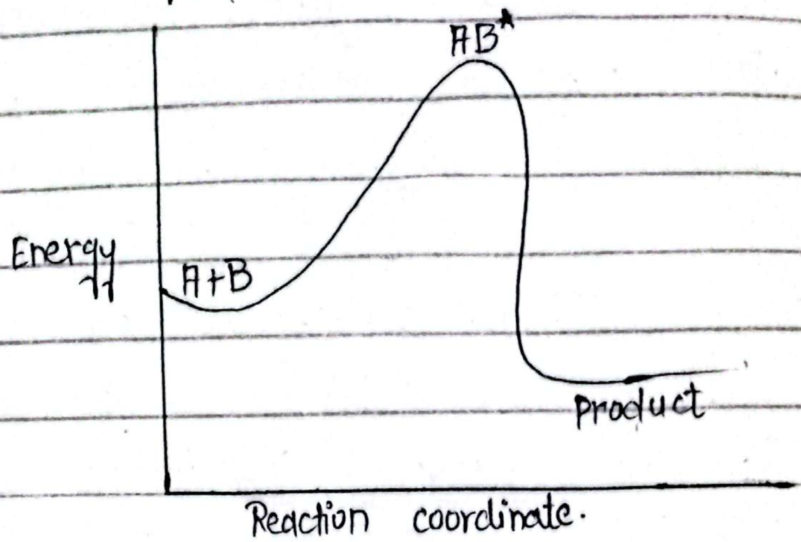
8- Rate of reaction is a function of vibrational frequency of activated complex.

Mathematically :-

The reaction species A and B can be represented as,

$$A + B \underset{}{\overset{K\neq}{\rightleftharpoons}} AB^{*} \xrightarrow{K\neq} P \quad \text{———} (1)$$

The energy profile for reaction eq (1)



⇒ Energy profile for the reaction $A+B \xrightarrow{K} P$.

The equilibrium constant for the first part of the reaction eq (1) is given by

$$K = \frac{[AB^{\neq}]}{[A][B]}$$

$$[AB^{\neq}] = K\neq [A][B] \quad \text{———} (2)$$

The rate of formation of products depends upon the concentration of the activated complex i.e,

$$\frac{d[P]}{dt} \propto [AB^{\neq}]$$

$$\frac{d[P]}{dt} = K\neq [AB^{\neq}] \quad \text{———} (3)$$

Where K is vibrational frequency of

AB.

According to statistical thermodynamics, the vibrational frequency of activated complex can be written as

$$k_{\neq} = \frac{k_B T}{h}$$

Where "k_B" is Boltzmann constant (1.381×10^{-23} J/k).

"T" is the absolute temperature in kelvin (k) and "h" is Plank's constant (6.626×10^{-34} Js).

Putting this value of $k_{\neq}$ in eq ③

$$\frac{d[P]}{dt} = \frac{k_B T}{h} [AB^{\ddagger}]$$

Putting the value of $[AB^{\ddagger}]$ from eq ②

$$\frac{d[P]}{dt} = \frac{k_B T}{h} K_{\neq} [A][B] \quad \text{——— ④}$$

Eq ④ represents the rate of a bimolecular reaction in which the reactants transform into products after passing through a transition state.

Whereas, the rate of a simple bimolecular reaction in which two reactants A and B combine to give the product P is given by

$$\frac{d[P]}{dt} = k_2 [A][B] \quad \text{——— (5)}$$

Where k_2 is the bimolecular rate constant for this simple reaction.

Comparing eq (4) and (5), we get

$$k_2 = \frac{k_B T}{h} K^{\neq} \quad \text{——— (6)}$$

As we know that the equilibrium constant of a reaction is related with the change in Gibbs free energy according to following expression.

$$\Delta F^{\neq} = -RT \ln K^{\neq}$$

Where $\Delta F^{\neq}$ is the change in free energy of activation and is obtained by subtracting the free energy of the activated complex from that of the reactants.

$$\Delta F^{\neq} = -RT \ln K^{\neq}.$$

Taking antilogarithm on b.s of the above eq, we get

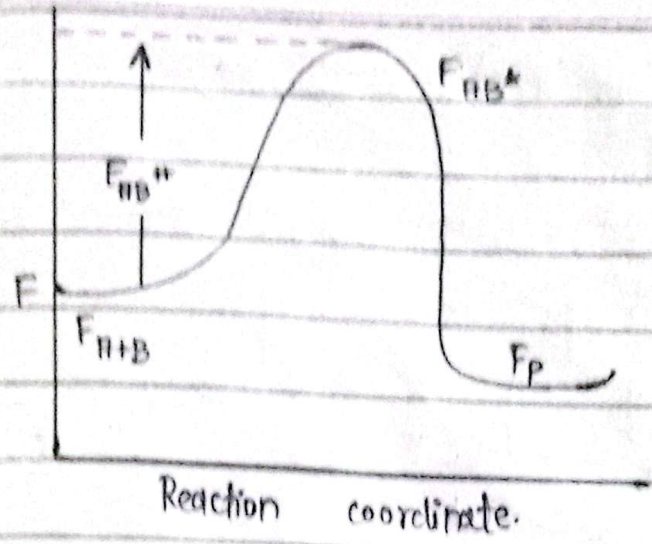
$$e^{-\Delta F^{\neq}/RT} = K^{\neq}$$

Putting this value of k in eq (6)

$$k_2 = \frac{k_B T}{h} e^{-\Delta G^{\neq}/RT} \quad \text{——— (7)}$$

The energy diagram for a general chemical reaction of the type

$$A + B \xrightarrow{k_2} P$$



According to the fundamental relations of thermodynamics, we know that

$$\Delta F^{\ddagger} = \Delta H^{\ddagger} - T\Delta S^{\ddagger}$$

where $\Delta F^{\ddagger}$, $\Delta H^{\ddagger}$ and $\Delta S^{\ddagger}$ represent the free energy, enthalpy and entropy of activation respectively, and "T" is the absolute temperature.

Putting this value of $\Delta F^{\ddagger}$ in eq (7), we get

$$k_2 = \frac{k_B T}{h} e^{-(\Delta H^{\ddagger} - T\Delta S^{\ddagger})/RT}$$

$$k_2 = \frac{k_B T}{h} e^{\Delta S^{\ddagger}/R} \cdot e^{-\Delta H^{\ddagger}/RT} \quad \text{——— (8)}$$

Eq (8) is known as Eyring eq.

$$\frac{k_2}{T} = \frac{k_B}{h} e^{\Delta S^{\ddagger}/R} \cdot e^{\Delta H^{\ddagger}/RT}$$

Taking ln on b.s of above eq, we get,

$$\ln \frac{k_2}{T} = \ln \frac{k_B}{h} + \frac{\Delta S^{\ddagger}}{R} - \frac{\Delta H^{\ddagger}}{RT} \quad \text{--- (9)}$$

This is the equation of straight line in intercept form where $\ln(k_2/T)$ is the dependant variable and $1/T$ is the independent variable. Plotting $\ln(k_2/T)$ versus $1/T$ gives a straight line with negative slope.

$$\text{Slope} = \frac{\Delta H^{\ddagger}}{R}$$

Hence $\Delta H^{\ddagger}$ can be calculated from slope as

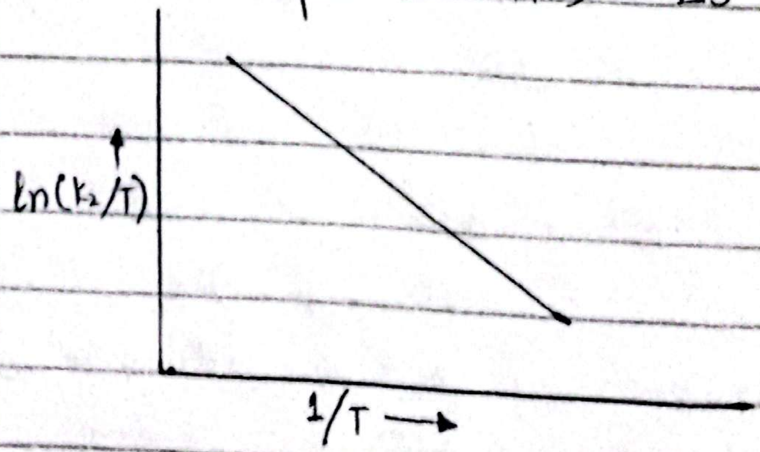
$$\Delta H^{\ddagger} = -R \times \text{slope}.$$

The intercept of the plot is

$$\text{Intercept} = \ln \frac{k_B}{h} + \frac{\Delta S^{\ddagger}}{R}$$

Rearranging above equation, the entropy of activation can be obtained as,

$$R\left[\text{Intercept} - \ln k_B/h\right] = \Delta S^{\ddagger}.$$



In given graph above Eyring plot is showing the variation in the value of $\ln(k_2/T)$ with $1/T$.

A COMPARISON OF TRANSITION STATE THEORY and ARRHENIUS THEORY:

Taking ln of eq (8), we get

$$\ln k_2 = \ln \frac{k_B}{h} - \ln \frac{1}{T} + \frac{\Delta S^{\neq}}{R} - \frac{\Delta H^{\neq}}{RT}$$

Differentiating the above eq with respect to $1/T$, we get

$$\frac{d \ln k_2}{d(1/T)} = 0 - \frac{1}{1/T} + 0 - \frac{\Delta H^{\neq}}{R}$$

$$\frac{d \ln k_2}{d(1/T)} = -T - \frac{\Delta H^{\neq}}{R}$$

$$-R \frac{d \ln k_2}{d(1/T)} = RT + \Delta H^{\neq} \quad \text{——— (10)}$$

Comparing eq (10) with Arrhenius eq

$$\left(E_a = -R \frac{d \ln k}{d(1/T)}\right)$$

$$E_a = RT + \Delta H^{\neq}$$

$$\Delta H^{\neq} = E_a - RT \quad \text{——— (11)}$$

Putting the value of $\Delta H^{\neq}$ in eq (8)

$$k_2 = \frac{k_B T}{h} e^{\Delta S^{\neq}/R} e^{-(E_a - RT)/RT}$$

$$k_2 = \frac{k_B T}{h} e^{\Delta S^{\neq}/R} e \, e^{-E_a/RT} \quad \text{——— (12)}$$

Comparing eq. (12) with Arrhenius eq.

$$\left(k = A e^{-E_a/RT}\right)$$

$$A = \frac{k_B T}{h} e^{\Delta S^{\neq}/R} e.$$

The above eq. indicates that the pre-exponential factor in Eyring eq. i.e. a function of Temperature.

3-EFFECT OF TEMPERATURE ON THE RATE OF A REACTION :

The rate constant of a reaction is a function of temperature. Any factor effecting the rate constant also effects the rate of a reaction. The rate of a reaction is therefore, also dependent upon temperature. As the temp. of the reaction mixture increases, the kinetic energy of the molecules also increases. Due to higher kinetic energy, there will be greater number of effective collisions and as a result the reaction will take place at a faster rate.

In 1889 Arrhenius experimentally observed the variation of rate constant 'k' with absolute temp. 'T'. He gave an emperical relationship between rate constant 'k' and temp. T.

$$\ln k = A' - \frac{B}{T} \quad \text{——} (1)$$

'A' and 'B' are the constants which depend upon the nature of the chemical reactions. According to eq (1) the increase of temp. T, increases the rate constant 'K'.

Eq (1) is the eq. of straight line $(y = -mx + c)$ and if the graph is plotted between $1/T$ on x-axis and log k on y-axis, a straight line is obtained with the negative slop. The value of 'B' is obtained from the slope of the straight line and that of Factor 'A' from the intercept of the

straight line.
eq ① is in the form of log and it can be converted into exponential form.

$$\ln k = A' - \frac{B}{T}.$$

Taking anti ln

$$k = e^{(A' - B/T)}$$

$$k = e^{A'} \times e^{-B/T}$$

$$k = Ae^{-B/T} \quad \text{——— ②}$$

Eq ② is the exponential form of the Arrhenius eq. and it serves the same purpose as eq ①. Anyhow, the factor 'B' was replaced by E_a/R and the accepted form of Arrhenius equation.

$$k = Ae^{-E_a/RT} \quad \text{——— ③}$$

$e^{-E_a/RT}$ is called Baltzmann factor. Its value is controlled by energy of activation E_a and the temperature 'T'. Both 'A' and 'E_a' are independent of temperature, and are determined by the properties of reacting molecules. These two factors have a great theoretical significance.

Taking ln on b.s of eq (3)

$$\ln k = \ln (Ae^{-Ea/RT})$$

$$\ln k = \ln A + \ln e^{-Ea/RT}$$

$$\ln k = \ln A - \frac{Ea}{RT} \ln e.$$

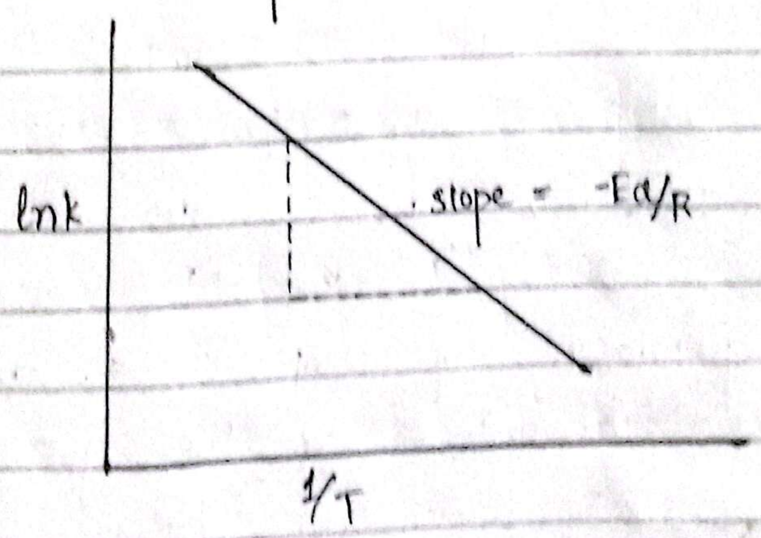
$\therefore \ln e = 1.$

$$\ln k = \frac{-Ea}{RT} + \ln A \quad \text{--- (4)}.$$

Eq (4) is the linear form of Arrhenius equation.

Following results can be derived from this equation,

1- This is the eq. of straight line in intercept form where $\ln k$ is the dependent variable and $1/T$ is the independent variable.



2- A plot of $\ln k$ versus $1/T$ gives a straight line. The slope and intercept of which can be used to calculate the Arrhenius parameters.

$$\text{Slope} = \frac{-E_a}{R}$$

$$E_a = -\text{slope} \times R$$

$$\text{Intercept} = \ln A$$

$$A = e^{\text{intercept}}$$

3- Let at temperature T_1, the rate constant for a reaction is k_1, hence the eq (4) will becomes:

$$\ln k_1 = \frac{-E_a}{RT_1} + \ln A \quad \text{—— (5)}$$

Similarly, at temperature T_2, the rate constant will be k_2, hence eq (4) will be:

$$\ln k_2 = \frac{-E_a}{RT_2} + \ln A \quad \text{—— (6)}$$

Subtracting eq (5) and (6), we get

$$\ln k_2 - \ln k_1 = \frac{E_a}{RT_2} + \frac{E_a}{RT_1}.$$

$$\ln \frac{k_2}{k_1} = \frac{E_a}{R}\left(\frac{1}{T_1} - \frac{1}{T_2}\right)$$

$$\ln \frac{k_2}{k_1} = \frac{E_a}{R}\left(\frac{T_2 - T_1}{T_1T_2}\right) \quad \text{--- (7).}$$

Eq (7) is another form of Arrhenius equation that is useful for calculating the value of Activation energy of a reaction when the value of rate constant at two different temperature is known for that reaction. In comparison to this, the graphical method of finding activation energy by using eq (4) is a time consuming process as one has to study the kinetics of a reaction at at least 4 or 5 different temperatures.

4- Differentiating eq (4) with respect to 'T', we get

$$\frac{d}{dT}(\ln k) = \frac{d}{dT}\left[\frac{-E_a}{RT} + \ln A\right]$$

$$\frac{d \ln k}{dT} = \frac{-E_a}{R}\left(\frac{d}{dT} T^{-1}\right) + 0$$

$$\frac{d \ln k}{dT} = \frac{-E_a}{R}(-1)T^{-1-1}$$

$$\frac{d \ln k}{dT} = E_a / RT^2 \quad \text{--- (8).}$$

eq (8) is the differential form of Arrhenius equation and can be used for the determination of activation energy as

$$Ea = RT^2 \frac{d\ln k}{dT}$$

5- Differentiating eq (4) with respect to 1/T, we get

$$\frac{d}{d(1/T)}(\ln k) = \frac{d}{d(1/T)}\left[\frac{-Ea}{RT} + \ln A\right]$$

$$\frac{d\ln k}{d(1/T)} = \frac{-Ea}{R}\left[\frac{d}{d(1/T)}\cdot\frac{1}{T}\right] + \frac{d}{d(1/T)}\ln A$$

$$\frac{d\ln k}{d(1/T)} = \frac{-Ea}{R}(1) + 0$$

$$Ea = -R\,\frac{d\ln k}{d(1/T)} \quad \text{——— (9)}$$

Where $d\ln k/d(1/T)$ is the slop of the plot of $\ln k$ versus $1/T$ and R is the general gas constant. Hence, determining the slope and putting this value in eq (9), enables us to calculate the value of activation energy. If, for a reaction, the plot of $\ln k$ versus $1/T$ does not give a straight line, the reaction is said to show a non-Arrhenius behaviour.

UNIMOLECULAR GAS PHASE REACTION:

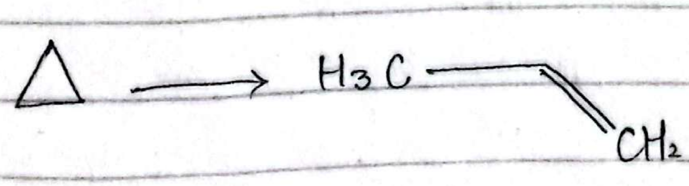
In such type of reactions, the reactants as well as the product exist in gaseous state. In these reactions, only one molecule is involved in the rate determining step and the order of reaction and decomposition reactions.

Decomposition of ethane and also that of ethyl iodide involves unimolecular gas phase reaction.

$$CH_3 - CH_3 \longrightarrow 2CH_3^{\bullet}$$

$$CH_3CH_2 - I \longrightarrow C_2H_4 + HI.$$

Another such example is the isomerization of cyclopropane.



⇒ KINETICS OF UNIMOLECULAR GAS PHASE REACTIONS:

Let us consider a reaction in

which a reactant A transforms into the product B after passing through a series of reactions.

$$A \longrightarrow B \quad \text{---} (1)$$

According to Lindmann Theory

$$A + A \underset{k_{-1}\,(\text{deactivation})}{\overset{k_1\,(\text{activation})}{\rightleftharpoons}} A + A^*$$

$$A^* \xrightarrow[\text{decomposition}]{k_2} B$$

$$A + B \underset{k_{-3}}{\overset{k_3}{\rightleftharpoons}} A^* + B$$

This mechanism reveals that the reactant A transforms into the product B after passing through an intermediate form i.e, A^*. The first step shows a reversible reaction involving the activation of 'A' to form an activated complex A^* and the deactivation of this complex to give back to reactant again. In the second step, activated complex decomposes to form the product B which then activates the molecules of the reactant to form the activated complex in the third step. The 2nd step is the slowest step of all is therefore, rate determining step.

$$\frac{d[B]}{dt} = k_2[A^*] \quad \text{—— (2)}$$

The concentration of activated complex A^* can be determined by steady state approximation.

$$\frac{d[A^*]}{dt} = k_1[A]^2 - k_{-1}[A^*][A] - k_2[A^*] + k_3[A][B] - k_3[A^*][B]$$

Applying steady state approximation, we get

$$k_1[A]^2 - k_{-1}[A^*][A] - k_2[A^*] + k_3[A][B] - k_{-3}[A^*][B] = 0$$

$$[A^*]\{k_{-1}[A] + k_2 + k_{-3}[B] = k_1[A]^2 + k_3[A][B]$$

$$A^* = \frac{k_1[A]^2 + k_3[A][B]}{k_1[A] + k_2 + k_{-3}[B]}$$

Putting value of $[A^*]$ in eq (2)

$$\frac{d[B]}{dt} = \frac{k_2\{k_1[A]^2 + k_3[A][B]\}}{k_{-1}[A] + k_2 + k_{-3}[B]} \quad \text{—— (3)}$$

Let the conc of A and B be 'a' and '0' initially and after a certain time 't' the conc change into $a-x$ and x respectively.

Hence, putting the values of the available conc of A and B in eq (3), we get

$$\frac{dx}{dt} = \frac{k_2\{k_1(a-x)^2 + k_3(a-x)(x)\}}{k_{-1}(a-x) + k_2 + k_{-3}(x)} \quad \text{——— (4)}$$

Since forward reactions of both the 1st and the 3rd step involve the activation of A while the backward reactions involves the deactivation of A^*. Hence k_3 can be assumed to be equal to k_1 and k_3 be equal to k_{-1}, therefore replacing k_3 by k_1 and k_{-3} by k_{-1} in eq (4), we get

$$\frac{dx}{dt} = \frac{k_2\{k_1(a-x)^2 + k_1(a-x)(x)\}}{k_{-1}(a-x) + k_2 + k_{-1}(x)}$$

$$\frac{dx}{dt} = \frac{k_2(k_1a^2 + k_1x^2 - 2k_1ax + k_1xa - k_1x^2)}{k_{-1}a - k_{-1}x + k_2 + k_{-1}x}$$

$$\frac{dx}{dt} = \frac{k_2(k_1a^2 - k_1ax)}{k_1a + k_2}$$

$$\frac{dx}{dt} = \frac{k_1k_2a(a-x)}{k_{-1}a + k_2} \quad \text{——— (5)}$$

let $$\frac{k_1k_2a}{k_{-1}a + k_2} = k' \quad \text{——— (6)}$$

Eq (5) will become

$$\frac{dx}{dt} = k'(a-x)$$

Hence, animolecalar gas phase

reactions are first order reactions and the first order rate constant k' depends upon the initial pressure of gas.

For small value of 'a' i.e when the pressure is low, then $k_{-1}a + k_2 \approx k_2$, applying this condition to eq (5)

$$k' = \frac{k_1 k_2 a}{k_2}$$

$$k' = k_1 a$$

The above eq shows that rate constant is directly proportional to a. at low pressure.

Hence, increasing the conc. of A increases the rate of the reaction and vice versa.

For large values of 'a' i.e, when the pressure is high, then $k_{-1}a + k_2 \approx k_{-1}a$, hence eq (6) will become

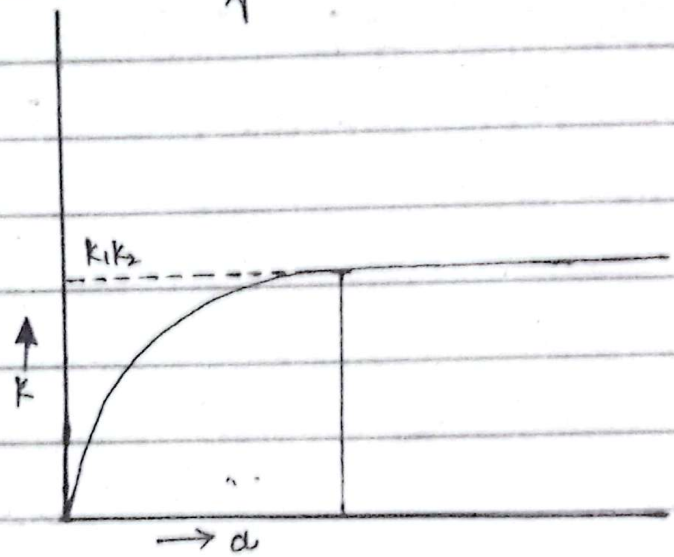
$$k' = \frac{k_1 k_2 a}{k_{-1} a}$$

$$k' = \frac{k_1 k_2}{k_{-1}}$$

As we know that $(k_1/k_{-1}) = K_1$,

where k_1 is the equilibrium constant for first step, the above eq. will become $k' = k_1 k_2$

Hence, at high pressure, the rate of the reaction is independent of 'a'. The variation in the value of k' with the change in the initial conc. (pressure) of the gas.



Taking reciprocal on b.s of eq (6)

$$\frac{1}{k'} = \frac{k_{-1}a + k_2}{k_1 k_2 a}$$

$$\frac{1}{k'} = \frac{k_{-1}}{k_1 k_2} + \frac{1}{k_1 a}$$

This is the equation of straight line in intercept form, where $1/a$ is the independent varible and $1/k'$ is the dependent variable.

k_1 can be calculated by taking the seciprocal of the slope of the plot $1/\alpha$ versus $1/k'$. Dividing the intercept by the slope gives the value of Hinshelwood constant i.e k_{-1}/k_2.

UNIMOLECULAR GAS PHASE REACTIONS IN THE PRESENCE OF AN INERT GAS:

Lindmann proposed the mechanism for reaction in the presence of an inert gas M.

$$A + A \underset{k_{-1}}{\overset{k_1}{\rightleftharpoons}} A^* + A,$$

$$A^* \xrightarrow{k_2} B$$

$$A + B \underset{k_{-3}}{\overset{k_3}{\rightleftharpoons}} A^* + B$$

$$A + M \underset{k_{-4}}{\overset{k_4}{\rightleftharpoons}} A^* + M.$$

The rate of reaction in terms of the rate of formation of the product B is given by

$$\frac{dB}{dt} = k_2 [A^*] \quad \text{——— (1)}$$

A^* is the reaction intermediate and its concentration cannot be determined experimentally, Hence, steady state approximation is used for this purpose.

$$\frac{d[A^*]}{dt} = k_1[A]^2 - k_{-1}[A^*][A] - k_2[A^*] + k_3[A][B] - k_{-3}[A^*][B] + k_4[A][M] - k_{-4}[A^*][M].$$

Applying steady state approximation, we get

$$k_1[A]^2 - k_{-1}[A^*][A] - k_2[A^*] + k_3[A][B] - k_{-3}[A^*][B] + k_4[A][M] - k_{-4}[A^*][M] = 0$$

$$k_{-1}[A^*][A] + k_2[A^*] + k_{-3}[A^*][B] + k_{-4}[A^*][M] = k_1[A]^2 + k_3[A][B] + k_4[A][M].$$

$$A^*\{k_{-1}[A] + k_2 + k_{-3}[B] + k_{-4}[M]\} = k_1[A]^2 + k_3[A][B] + k_4[A][M].$$

$$[A^*] = \frac{k_1[A]^2 + k_3[A][B] + k_4[A][M]}{k_{-1}[A] + k_2 + k_{-3}[B] + k_{-4}[M]}$$

Putting the value of A^* in eq

$$\frac{d[B]}{dt} = \frac{k_2\{k_1[A]^2 + k_3[A][B] + k_4[A][M]}{k_{-1}[A] + k_2 + k_{-3}[B] + k_{-4}[M]} \longrightarrow (8)$$

Let the conc of A and B be 'a' and '0' initially and after a certain time 't' the conc. change into $a-x$ & x respectively.

Putting in eq (8)

$$\frac{dx}{dt} = \frac{k_2\{k_1(a-x)^2 + k_3(a-x)(x) + k_4(a-x)[M]\}}{k_{-1}(a-x) + k_2 + k_{-3}(x) + k_{-4}[M]}$$

$$\frac{dx}{dt} = \frac{k_2(a-x)\{k_1(a-x) + k_3(x) + k_4[M]\}}{k_{-1}(a-x) + k_2 + k_{-3}(x) + k_{-4}[M]}$$

Replacing k_3 by k_1 and k_{-3} by k_{-1} in above eq

$$\frac{dx}{dt} = \frac{k_2(a-x)\{k_1(a-x) + k_1(x) + k_4[M]\}}{k_1(a-x) + k_2 + k_{-1}(x) + k_{-4}[M]}$$

$$\frac{dx}{dt} = \frac{k_2(a-x)(k_1 a - \cancel{k_1 x} + \cancel{k_1 x} + k_4[M])}{k_{-1}a - \cancel{k_{-1}x} + k_2 + \cancel{k_{-1}x} + k_{-4}[M]}$$

$$\frac{dx}{dt} = \frac{k_2(k_1 a + k_4[M])(a-x)}{k_{-1}a + k_2 + k_{-4}[M]} \quad \text{--- (9)}$$

Let $$\frac{k_2(k_1 a + k_4[M])}{k_{-1}a + k_2 + k_{-4}[M]} = k'' \quad \text{--- (10)}$$

Hence eq (9) becomes

$$\frac{dx}{dt} = k''(a-x)$$

The above eq shows that the reaction is first order. k'' depends on the conc. (pressure) of M as well as the initial conc. of A.

4- BIMOLECULAR REACTION IN SOLUTION: (IONIZATION REACTIONS)

Solution phase reactions may be classified into diffusion controlled and activation controlled reactions depending upon whether the diffusion or activation step serves as the rate limiting step. Another basis of classification is the nature of reaction i.e, whether the reaction involves transfer of electron namely redox reactions or exchange of protons /hydroxyl ions namely acid-base reactions. The reactions in solution are very fast as compared to those in gas phase due to the solvent cage effect. Molecules of different reactants are in a cage of the molecules of solvent and collide with each other many times and are converted into products in very short interval of time but in gas phase reaction the probability of collision of molecules of reactants is less as compared to that in solution.

DIFFUSION CONTROLLED AND ACTIVATION CONTROLLED REACTIONS:

Consider a bimolecular reaction,

$$A + B \xrightarrow{k_2} P \text{ ——— } (1)$$

In which reactants approach each other by diffusion and form an encounter pair AB in the presence of a solvent.
The rate of such a reaction is given by:

$$\frac{d(P)}{dt} = k_2 (A)(B) \text{ ——— } (2)$$

Reaction (2) is supposed to involve following mechanism

$$\left.\begin{array}{l} A + B \xrightarrow[\text{Diffusion}]{k_1} AB \\ \\ AB \xrightarrow{k_{-1}} A + B \\ AB \xrightarrow[\text{Activation}]{k_1'} P \end{array}\right\} \text{ ——— } (3)$$

Very first step of mechanism involves the formation of an encounter pair AB which then, following two different paths, decomposes to give the reactant again in 2nd step and transform into product in the 3rd step.

The third step of the mechanism is the rate determining step as it is the slowest of all. Hence, the rate of the reaction in terms of the formation of the product 'P' is given by

$$\frac{d[P]}{dt} = k_1'[AB] \quad \text{——} \; (4)$$

The expression for the rate of formation of AB can be written as

$$\frac{d[AB]}{dt} = k_1[A][B] - k_{-1}[AB] - k_1'[AB]$$

Applying steady state approximation, we get:

$$k_1[A][B] - k_{-1}[AB] - k_1'[AB] = 0$$

$$[AB](k_{-1} + k_1') = k_1[A][B]$$

$$[AB] = \frac{k_1[A][B]}{k_{-1} + k_1'}$$

Putting the value of [AB] in eq (4)

$$\frac{d[P]}{dt} = \frac{k_1' k_1 [A][B]}{k_{-1} + k_1'} \quad \text{——} \; (5)$$

Comparing eq (2) and eq (5) the value of k_2 comes out to be

$$k_2 = \frac{k_1 k_1'}{k_{-1} + k_1'} \quad \text{—— (6)}$$

If k_1' is very much greater than k_{-1}, then the denominator of the above eq. will reduce to k_1', apply this condition to eq (6)

$$k_2 = k_1 \quad \text{—— (7)}$$

Eq. (7) shows that k_2 depends on the diffusion step of mechanism eq. (3). Therefore, diffusion is the rate controlling step under this condition. Such reactions are called diffusion reactions.

If k_1 is very much smaller than k_{-1}, then the denominator in the eq (6) will reduce to k_{-1},

$$k_2 = \frac{k_1 k_1'}{k_{-1}} = k_1 k_1' \quad \text{—— (8)}$$

The eq. (8) shows the dependence of k_2 on the rate constant for third step of the mechanism. This reaction involves the formation of product that is associated with the certain amount of activation energy and is therefore said to be activation controlled reaction. According to transition state theory, the process of an activation controlled reaction depends on the following factors.

★ Solvent Effect:

The effect of solvent on the rate of a reaction can be explained in terms of the entropy of activation using transition state theory. The relation between k_2 and $\Delta S^{\ddagger}$ is:

$k_2 = \frac{k_B T}{h} e^{\Delta S^{\ddagger}/R} e^{-\Delta H^{\ddagger}/RT}$. According to this relation, the positive value of $\Delta S^{\ddagger}$ favours the reaction whereas the negative value decreases the value of k_2. Moreover, an increase in the positive values $\Delta S^{\ddagger}$ increases the rate of reaction while an increase in the negative values of $\Delta S^{\ddagger}$ decreases the value of k_2. The value of $\Delta S^{\ddagger}$ is purely associated with the nature of solvent. For example, the formation of activated complex from the reactants having similar charges is accompained by a decrease in the entropy of activation and hence the rate of reaction decreases. The entropy decreases due to fact that the magnitude of the charge on the activated complex is greater than that on the reactants due to which a greater number of molar molecules of the solvent get arranged, surrounding the molecules of the activated complex. Hence, the degree of randomness decreases, decreasing the entropy of activation and lowering the reaction rate. In case of the reactants with opposite charges, the magnitude of the charge on the activated complex, so formed, with be less, offering less home for the polar molecules of the solvent and promoting disorderness due to which the entropy of activation increases and hence, the rate of reaction increases. The entropy of a reaction between two neutral species in aqueous medium depends upon the difference in the polarity of the activated complex and that of the reactants. If the activated complex has more polarity than the reactants, then the entropy of activation will decrease, decreasing the rate of reaction, whereas the reactions with the reactants and the activated complex of same polarity proceed without any change in the entropy of activation.

5- FAST REACTIONS AND THEIR METHODS OF STUDY:

Fast reactions are very high speed reactions and therefore, exhibit very high values of rate constants and very short half-life periods. Such reactions may be examplified by the solution phase reactions involving ionic species.

e.g

$$NaOH + HCl \longrightarrow NaCl + H_2O.$$

1- FLOW METHODS:

These methods were introduced by Roughton and Hartridge in 1923 in order to study the kinetics of those reactions whose half-life period is even less than a second. Here, reactants are mixed within a very short time span, that is why these methods are known as flow methods.

In flow, methods, the reactants, through two separate driving syringes are mixed thouroghly in a mixing chamber and this mixture is allowed to flow through an observation tube where the concentration of the components is monitored by some physical technique like optical, electrical and thermal methods of analysis. Flow methods have been classified as continuous flow method and stopped flow method depending upon whether the velocity of flowing mixture is kept constant or the observation of reaction mixture is made at some fixed distance.

In this conventional flow technique, the reaction mixture is allowed to flow through an observation tube at a constant velocity, while the reaction goes on during the flow. The determination of the composition of the reaction mixture at various points along the observation tube by a moveable analytical instrument corresponds to the determination of the rate of reaction at different time interval. Let the velocity of the mixture and distance between the points of observation and mixing chamber are 'v' and 's' respectively then v/s gives the value of the time for which the mixture has flown and reaction has progressed which helps to determine rate of reaction. The drawback of this method is the need of a very large volume of the reaction mixture so as to spread it along the whole tube rapidly.

In stopped flow method, the mixing chamber is provided with a syringe insted of a flow tube, which stops the flow of the mixture at a fixed point while the reaction goes on which is then monitored as a function of time using some analytical technique. This method has many advantages over continuous flow methods. A very small amount of mixture is enough for analysis. Moreover, we can have a wide record of progress of reaction with no wastage of reactants by stopped flow method.

2- PULSE METHODS:

Pulse methods are employed for the kinetic study of very fast reversible reactions. In these methods, the reaction system is allowed to attain equilibrium i.e, the concentration of the reactants as well as the products become constant at this stage. The system is then perturbed by varing any of the external factors. Eigen in 1950 introduced a pulse method called relaxation technique to study fast reactions and got Noble Prize for his work in 1960.

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